HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE413: Computer Graphics

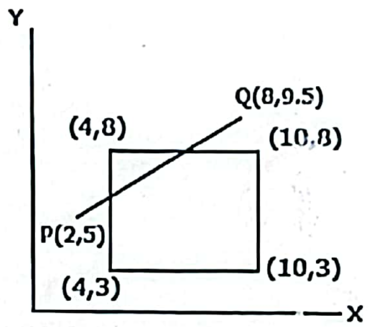
Batch: 18th | Semester: Spring 2023

Time: 2 Hours

Date: 05/09/2023 **Final Examination** Total Marks: 40

Answer any four questions of the followings. $4 \times 10 = 40$

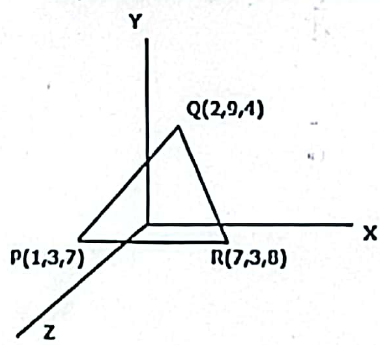
1. (a) Explain polygon surface table with example. 3
   (b) Describe the binary region coding system in Cohen-Sutherland Line clipping algorithm. 3
   (c) Apply Cohen-Sutherland Line clipping algorithm to clip the line (P,Q) of the following figure: 4



2. (a) Draw the 3D transformation pipeline. 2
   (b) Define color model in computer graphics. Explain RGB color model. 4
   (c) Describe CMY color model. If the coordinates of a color is given (0.85, 0.78, 0.50) in YIQ color model, convert the relevant coordinates in CMY color model. 4

3. (a) What is visible surface detection algorithm? 2
   (b) State the **Z-Buffer** algorithm to find the depth values for visible surfaces. 4
   (c) Define **BSP** tree. How does BST tree involves in visibility testing? Give an example. 4

4. (a) Explain 3D reflection technique with matrix representation. 3
   (b) If $S_x = 1.2, S_y = 0.8$ *and* $S_z = 1.1$, find the new coordinates for the following triangle: 3



   (c) Given a homogeneous point (4, 7, 3). Apply rotation 45 degree towards X and Z axis and find out the new coordinate points. 4

5. (a) Define projection in computer graphics. Explain its types. 5
   (b) Describe the types of Quadric Surface with proper equation of each. 5

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 417 : Compiler Design

Batch: 18th | Semester: Spring 2023
Time: 2 Hours
Date: 09/09/2023 | **Final Examination** | Total Marks: 40

Answer any **five** questions: $5 \times 8 = 40$

1. (a) Explain the structure of a Compiler. — 3
   (b) What are the steps of a Language Processing System? Briefly describe. — 5

2. (a) What is the role of a parser? — 4
   (b) Briefly explain different error recovery strategies in Compiler Design. — 4

3. (a) What is Context Free Grammar? — 2
   (b) Show the step by step procedure of constructing a Parse Tree (Left-most Derivation) for the expression **id – id / id**, where, — 6

   Production rules:

   $E \rightarrow E - E$

   $E \rightarrow E / E$

   $E \rightarrow id$

4. (a) Find out if the following Strings belong to the given Grammar- — 4

   i. abbcbba ii. baacaab iii. abcab iv. aacbb

   Production rules:

   $S \rightarrow aSa \quad S \rightarrow bSb \quad S \rightarrow c$

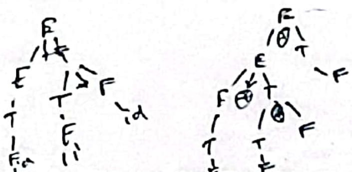
   (b) What is Attribute Grammar? Briefly describe Synthesized Attribute and Inherited Attribute. — 4

5. (a) What is Left Recursion in Context Free Grammar? How can you remove it? — 3
   (b) Eliminate Left Recursion from the following Grammar: — 5

   $E \rightarrow E + T \mid T$

   $T \rightarrow T * F \mid F$

   $F \rightarrow id$



6. (a) What do you mean by the Semantic Analysis phase in Compiler Design? — 3
   (b) Mention and describe some of the semantic errors that the semantic analyzer is expected to recognize. — 5

7. (a) What are the two common representations for Intermediate Code in Compiler Design? — 2
   (b) What is the Three Address Code? How would you represent the expression **'a = b + c * d'** in three-address code? Provide the corresponding quadruple and triple representations for this expression. — 6

HAMDARD UNIVERSITY BANGLADESH

FACULTY OF BUSINESS ADMINISTRATION

MGT 419/MGT 417/BBA 321 Industrial Operational Management/Operations Management

Semester: Spring 2023

11/09/2023 Final Examination Time: 2 Hours

Total Marks: 10x4=40

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answers to the questions should be neat and clean.
- Answer any **four (4)** of the following questions.

01. a. What is job design? State the method of job analysis. 05

b. Illustrate the disadvantages of specialization in business. 05

02. a. List the requirements of successful team building. 03

b. An analyst has been asked to prepare an estimate of the proportion of time that a turret lathe operator specialist spends adjusting the machine, with an 90% confidence level. Based on previous experience, the analyst believes the proportion will be approximately 30%. 07

1) If the analyst uses a sample size of 400 observations, what is the maximum possible error that will be associated with the estimates?
2) What sample size would be analyst need in order to have the maximum error be no more than ±5 percent?

03. a. Compare between time-based and output-based systems. 03

b. A time study analyst wants to estimates the number of observation that will be needed to achieve a specified maximum error, with a confidence of 98 percent. A preliminary study yielded a mean of 6.2 minutes and a standard deviation of 1.1 minutes. Determine the total number of observations needed for these two cases: 07 $z = 2.33$

1) A maximum error of ±6% of the sample mean. $n = \left(\frac{zs}{ax}\right)^2$
2) A maximum error of .60 minutes. $n = \left(\frac{zs}{e}\right)^2$

04. a. Explain the elements of a good forecast. 05

b. Describe how to forecast based on time series data. 05

05

a. Discuss the business areas that uses forecasting associate with examples. **05**

b. Outline the principles of total quality management (TQM) **05**

06

a. Outline the functions of inventory management elucidate with examples. **03**

b. The Dine Corporation is both a user of brass couplings. The firm operates 200 days a year and uses the couplings at a steady rate of 50 per day. Couplings can be produced at a rate of 250 per day. Annual storage cost is \$2 H per coupling, and machine setup cost is \$60 per run. **07**

1) Determine the economic run quantity.

$$Q_0 = \sqrt{\frac{2DS}{H}} \times \sqrt{\frac{P}{P-U}}$$

2) Approximately how many runs per year there will be?

3) Compute the maximum inventory level.

4) Determine the length of the pure consumption proportion of the cycle.

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 411: Artificial Intelligence

Batch: 18th

Semester: Spring 2023

Date: 16/09/2023 **Final Examination** Time: 2 Hours

Total Marks: 40

Answer any five questions from the following: 5x8= 40

1 (a) What are the key steps and strategies for someone looking to become a recognized domain expert? 3

(b) Categorize the types of rules listed below into their respective classifications. 5×1=5

i. IF the oven timer goes off
AND the food smells cooked
THEN remove the dish from the oven

ii. IF the temperature drops below freezing point
THEN the water turns into ice

iii. IF you're stuck in traffic during rush hour
THEN consider taking an alternate route suggested by your GPS

iv. IF you have a job interview
AND it's a formal setting
THEN wear a suit and tie

v. IF you're trying to lose weight
AND you're feeling hungry between meals
THEN eat a healthy snack like fruits or nuts

2 (a) Illustrate the basic structure of a rule-based expert system. 3

(b) Design a model-based reflex agent. 5

3 (a) In the context of artificial intelligence, can expert systems, despite their knowledge and algorithms, still make mistakes? 3

(b) You are given a set of mathematical conditions and a database of facts. Your task is to determine whether a specific statement is true or false based on the following rules and database: 5

Database:

A is true.

B is false.

C is true.

D is false.

E is true.

Knowledge Base Rules:

If X and Y are both true, then Z is true: (X & Y -> Z)

If A is true, then X is true: (A -> X)

If B is false, then Y is true: (~B -> Y)

If C is true and D is false, then X is false: (C & ~D -> ~X)

If E is true, then Y is true: (E -> Y)

Is Z true or false based on the given database and knowledge base?

4 (a) What factors or considerations should guide the selection between forward and backward chaining when determining the appropriate inference method for a particular problem or scenario? 2

(b) Show the matching and firing cycles of the rules by forward chaining. 6

Database					
B	N	P	Q	R	Y
Knowledge Base					
P & Q & S -> Z					
B -> X					
Y -> S					
R & X -> W					
F & N -> H					

5 (a) Imagine a rational agent operating in a financial trading environment. What actions might it take to maximize its performance measure, and how would it use percept sequences to make decisions? 2

(b) What are the essential components of a PEAS framework for a Medical Diagnosis System? 3

(c) Imagine you're designing a video game with two distinct worlds: one static and the other semidynamic. Describe how these worlds would feel and behave differently for the players. What unique challenges and surprises might each world offer to the players' experience? 3

(a) How does CNN differ from traditional feedforward neural networks in terms of architecture and functionality? 2

(b) Explain the concept of feature learning in CNNs. How do convolutional layers extract meaningful features from input data, such as images? 3

(c) What is the significance of the term "stride" in convolutional layers of CNNs? How does the stride value affect the size of the output feature maps? 3

7 (a) Provide a comprehensive explanation of two primary components of Natural Language Processing (NLP)? 2

(b) Imagine you have the following sentence: "The quick brown fox jumps over the lazy dog." Using a Count Vectorizer, demonstrate how you would convert this sentence into a numerical representation. 2

(c) Briefly outline the stages of the NLP pipeline and how they work together. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 415: Communication Engineering

Batch: 18th
Date: 18/09/2023

Semester: Spring 2023
Time: 2 Hours
Total Marks: 40

Final Examination

Answer any five questions: $5 \times 8 = 40$

1. (a) What are the advantages of dividing an Ethernet LAN with a bridge? 4
   (b) List the tasks performed by the signaling system in a Telephone Network. 4

2. (a) Describe the SS7 services and its relation to the telephone network. 4
   (b) Draw and explain a STS-3 frame with its overhead. 4

3. (a) Illustrate the subfields in Frame Control (FC) field of MAC layers in IEEE 802.11 standard. 4
   (b) Draw and explain Wireless LAN (IEEE 802.11) architecture. 4

4. (a) Compare the medium of a wired LAN with that of a wireless LAN in today's communication environment. 4
   (b) Compare Piconet with Scatternet in the Bluetooth architecture. 4

5. (a) What is a Footprint? Explain the purposes of GPS. 4
   (b) Briefly explain the relationship between the Van Allen belts and Satellites. 4

6. (a) Write the functions of a mobile switching center. 4
   (b) Describe the function of the CDMA in IS-95. 4

7. (a) Which one has more overhead, a Hub or a Switch? Explain. 4
   (b) List the characteristics for get membership in a VLAN. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 435: Web Engineering

Batch: 18th

Semester: Spring 2023

Date: 20/09/2023

Final Examination

Time: 2 Hours
Total Marks: 40

Answer any **five** of the following questions. $5 \times 8 = 40$

1. (a) What is computer security and network security in web? 2
   (b) Explain Integrity and Confidentiality of web security. 6

2. (a) What is Layout Design? Write the basic steps of Layout Design. 2
   (b) Define the following terms with examples: $3 \times 2 = 6$
      i. Authentication
      ii. Authorization
      iii. Accountability

3. (a) Briefly explain the planning guidelines of web applications. 2
   (b) Write short notes for the following HTTP authentication protocols: $3 \times 2 = 6$
      i. Basic Authentication Protocol
      ii. Digest Authentication Protocol
      iii. Client Authentication Protocol

4. (a) Describe the web application project scopes. 3
   (b) What is Pair Walkthrough? Explain the mechanics of Pair Walkthrough. 5

5. (a) Write about the tools of a use-case diagram in a web application. 3
   (b) Consider **Ananda Shipping Management (ASM)** prides itself on having up-to-date information on the processing and current locations of each shipped item. To do this, ASM relies on a company-wide information system. **Shipped items** are the heart of the ASM product tracking information system. Shipped items may be of two types: goods and foods. They can be characterized by **item number (unique), weight, dimensions, insurance amount, destination, and final delivery date.** Shipped items are received into the BSM system at a single retail center. Shipped items make their way to their destination via one or more standard BSM transportation events (i.e. flights, truck deliveries). These transportation events are characterized by a **unique schedule Number, a type (e.g., flight, truck), and a delivery Route.** 5
   Design a use-case diagram to implement the scenario described above.

6. (a) Write the JavaScript Regular Expression code for the followings: $3 \times 1 = 3$
      i. Search for a word that begins with the pattern **'good'**
      ii. Search for a word that ends with the pattern **'noon'**
      iii. Search for a stand-alone word that is not begins and end with the pattern **'noon'**
   (b) Explain the generic capabilities should exist in a modeling language. 5

7. (a) Write the web application quality requirements tree structure. 4

(b) Write the PHP code to insert data in a database called 'myDB'. The form fields are user name, password, and mobile number with the database connection code. 4